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Monday

MLA0205 - Fundamentals of ML

Assignment - II

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Task 1 - Linear Regression:

Given $x = 1$

$y = 2.1, 3.9, 6.2, 7.8, 10.1$

using normal equations:

$$\omega_1 = \frac{5(110.2) - 15(30.1)}{5(55) - 225} = 1.99$$

$$\omega_2 = \frac{30.1 - 1.99(15)}{5} = 0.05$$

(a) regression equation is:

$$\boxed{y = 0.05 + 1.99x}$$

(b) Predict y at $x = 6$:

$$y = 0.05 + 1.99(6)$$

$$\boxed{y = 11.99}$$

(c) Calculate RSS, MSE and R^2:

$$RSS = \sum (y - \hat{y})^2$$

$$RSS = 0.107$$

$$MSE = \frac{RSS}{n} = \frac{0.107}{5} = 0.0214$$

$$R^2 = 1 - RSS/TSS = 0.9973$$

(d) Analyse the reliability of prediction:

The R^2 value is close 1, indicating that the model fits training data very well. However, predicting at $x = 6$ is extrapolation because the available data ranges from 1 to 5.

Task2: Ride Regression

Given $S_{xx} = 10$, $S_{xy} = 19.9$, $\lambda = 5$

The ride regression

$$w = \frac{S_{xy}}{S_{xx} + \lambda}$$

$$w = \frac{19.9}{10+5}$$

$$\boxed{w = 1.3267}$$

Least-squares slope:

$$w_{LS} = \frac{19.9}{10} = 1.99$$

$$\text{Percentage shrinkage} = \frac{1.99 - 1.3267}{1.99} \times 100$$

$$= 33.33\%$$

Bias-variance effect: Increasing λ increases bias and decreases variance. It reduces overfitting, but a very large λ may cause underfitting.

Task3: Naive Bayes Classiffication:

Given 10 emails:

Spam = 5, Ham = 5

(a) Calculate prior & conditional probabilities

Prior probabilities

$P(\text{spam}) = 5/10 = 0.5$

$P(\text{Ham}) = 5/10 = 0.5$

Conditional Prob

	Spam	Ham
P(offer=Y)	0.8	0.2
P(offer=N)	0.2	0.8
P(Link=Y)	0.8	0.2
P(Link=N)	0.2	0.8

(b) Classify offer = y, Link = y without smoothing

Using Naive Bayes:

$$P(\text{Spam}|X) = \frac{0.5\,(0.8)\,(0.8)}{0.5\,(0.8)(0.8) + 0.5\,(0.2)(0.2)}$$

$$= \frac{0.32}{0.34}$$

$$P(\text{Spam}|X) = 0.9412$$

∴ The email is classified as spam

(c) Repeat with Laplace smoothing (a=1)

Laplace smoothing formula $P = \frac{\text{Count} + \alpha}{\text{Total} + \alpha k}$

For spam:

$$P(\text{offer} = y|\text{spam}) = P(\text{Link} = y|\text{spam}) = 5/7$$

For Ham

$$P(\text{offer} = y|\text{Ham}) = P(\text{Link} = y|\text{Ham}) = 2/7$$

Posterior Prob.

$$P(\text{Spam}|X) = \frac{25}{25+4}$$

$$= 0.8621$$

∴ The email is classified as spam.

(d) Classify offer = N, Link = N with smoothing

$$P(\text{Spam}|X) = \frac{0.5\,(2/7)^2}{0.5\,(2/7)^2 + 0.5\,(5/7)^2}$$

$$P(\text{Spam}|X) = 0.1379$$

∴ The email is classified as Ham

Task 4: Logistic Regression:

Given

$$w_1 = 0, \quad w_2 = 0, \quad b = 0$$

Learning rate = 0.5

Samples: $x^{(1)} = (1,2)$, $y_1 = 1$

$x^{(2)} = (2,1)$, $y_2 = 0$

Step1: Using sigmoid function:

$$\sigma(z) = 1/1+e^{-z}$$

$$P_1 = P_2 = 0.5$$

Step2: For mean cross-entropy loss:

$$\partial L/\partial w = \frac{1}{n} \Sigma(P-y)\, x$$

gradients are $dw_1 = 0.25$

$dw_2 = -0.25$

Step3: Update parameters: $db = 0$

Gradient descent

$$w_{new} = w - \eta \nabla L$$

$$w_1 = 0 - 0.5(0.25) = -0.125$$

$$w_2 = 0 - 0.5(-0.25) = 0.125$$

$b = 0$

Step4: updated parameters => $w = (-0.125, 0.125)$

$b = 0$

Samp1 => $P_1 = \sigma(0.125) = 0.5312$

Sample2 => $P_2 = \sigma(-0.125) = 0.4688$

Step5: Initial loss

$L_{befor} = 0.6931$

After updating

$L_{after} = 0.6596$

Task 5: Decision Tree - Entropy & Information Gain

Given Total samples = 10

Yes = 6, No = 4

$A = a_1$: 4 Yes, 0 No

$A = a_2$: 2 Yes, 4 No

(a) Calculate entropy of the parent node:

$$H(S) = -P_1 \log_2 P_1 - P_2 \log_2 P_2$$

$$= -0.6 \log_2(0.6) - 0.4 \log_2(0.4)$$

$$\boxed{H(S) = 0.971}$$

(b) Calculate Information gain:

Entropy of first child $H(a_1) = 0$

Entropy of 2nd child $H(a_2) = -2/6 \log_2 2/6 - 4/6 \log_2 4/6$

$$H(a_2) = 0.9183$$

Weighted entropy:

$$= 0.4(0) + 0.6(0.9183) = 0.55098$$

$$IG = 0.971 - 0.55098$$

$$= 0.420$$

(c) Calculate Gini impurity decrease:

Parent Gini:

$$G(S) = 1 - (0.6^2 + 0.4^2) = 0.48$$

Child Gini: $G(a_1) = 0$

$$G(a_2) = 1 - (1/3)^2 - (2/3)^2 = 0.444$$

$$WCG = 0.4(0) + 0.6(0.444) = 0.2667$$

Gini decrease $= 0.48 - 0.2667$

$$= 0.2133$$

Updated

Task 6: Cost - Complexity Pruning:

Given subtree errors = 3
leaves = 4
single leaf errors = 8

using error rates:

$$0.03 + 4\alpha = 0.08 + \alpha$$
$$3\alpha = 0.05$$
$$\boxed{\alpha = 0.01667}$$

For $\alpha = 0.01$
subtree cost = 0.07
Leaf cost = 0.09
Do not prune

For $\alpha = 0.02$
Subtree cost = 0.11
Leaf cost = 0.10
Prune the subtree

Task 7: Neural Network

Given $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$

Forward pass:

$$h = \sigma(0.4 + 0.3(2))$$
$$h = 0.7311$$
$$\hat{y} = \sigma(0.5 \times 0.7311)$$
$$\boxed{\hat{y} = 0.5904}$$

Initial loss

$$E = \frac{1}{2}(1 - 0.5904)^2$$
$$\boxed{E = 0.0839}$$

Error terms

$$\delta_o = 0.09906, \quad \delta_h = 0.009738$$

Updated weights:

$$v = 0.5362$$

$$w_1 = 0.40487, \quad w_2 = 0.30974$$

New loss:

$$E_{new} = 0.08105$$

The loss decreases after backpropagation

Task 8: Support Vector Machine:

Given $w = (0.5, 0.5)$

$b = -2$

(a) Support Vectors:

$$f(3,3) = 1$$

$$f(1,1) = -1$$

both lie on margin hyperlanes

margin width: $2 / \sqrt{0.5^2 + 0.5^2} = 2.828$

(b) Classification:

(4,2): class +1, distance = 1.414

(1,2): class -1, distance = 0.707

(c) Hinge loss:

$$L = \max(0, 1 - y f(x))$$

* for (1,2) Loss = 0.5
* for (2,2) Loss = 1

∴ Both points violate the margin

Task 9 : Ensemble Learning - Majority voting:

Given Accuracy = 0.70

(a) Calculate accuracy for 3 classifiers using majority voting

$$P = \binom{3}{2}(0.7)^2(0.3) + (0.7)^3$$

$$= 3(0.49)(0.3) + 0.343$$

$$\boxed{P = 78.4\%}$$

(b) Calculate accuracy for 5 classifiers:

$$P = \sum_{k=3}^{5} \binom{5}{k} (0.7)^k (0.3)^{5-k}$$

$$= 0.3087 + 0.36015 + 0.16807$$

$$\boxed{P = 83.692\%}$$

(c) Analyze the result:

Adding more classifiers improves ensemble accuracy.

Task 10: AdaBoost

Given 5 samples, Initial weight = 0.2 each

Samples 2 and 5 are misclassified

weighted error

$$\epsilon = 0.2 + 0.2$$

$$= 0.4$$

Learner weight

$$\alpha = \frac{1}{2} \ln \frac{1 - 0.4}{0.4}$$

$$\alpha = 0.2027$$

Un normalised weights

=> Correct samples (1, 3, 4) = 0.1633 each

* wrong samples (2, 5) = 0.2449 each

Normalised weights:

Samples	Weight
1, 3, 4	0.1667 each
2, 5	0.2449 each

Misclassified samples receive higher weights, so the next learner focuses on them.

Task 11: Linear Discriminant Analysis

Given $\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$

(a) Discriminant direction:

$$w = \Sigma^{-1} (\mu_2 - \mu_1)$$

$$w = (1.4286, -1.7143)$$

Equal-prior threshold:

$$c = 0.2857$$

(b) For $x = (2.5, 1)$

$$w^T x = 1.8571$$

Since $1.8571 > 0.2857$ classification is ω_2

(c) Unequal priors:

$$c' = 0.2857 + \ln(0.8/0.2)$$

$$\boxed{c' = 1.6719}$$

Since $1.8571 > 1.6719$ classification remains ω_2.

Task 13: Classification Metrics and Cost

(a) Performance metrics

Metric	Model A	Model B	Model C
Accuracy	88%	80.5%	93%
Precision	0.4375	0.3208	0.6667
Recall	0.70	0.85	0.60
F1-score	0.5385	0.4659	0.6316

(b) Total expected cost

Formula => cost = FN (50000) + FP (5000)

Model	Total cost
A	₹ 19,50,000
B	₹ 16,50,000
C	₹ 21,50,000

(c) Best model

Model B is selected because it has lowest financial cost. Although Model C has the highest accuracy, it misses more difficulties

(d) Improvements:

1) Adjust the classification threshold to reduce false negatives.

2) Use cost-sensitive learning & class weighting.